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SPECIFICA TECHNIQUE

Gas measurement systems

Hourly gas measuring devices

Part 10: Security

UNI/TS 11291-10

February 2013

Gas measurement systems

Hourly based gas metering systems

Part 10: Safety

The technical specification defines the requirements for the security of devices used in the remote management network and for the security of data exchanged between the devices themselves. The technical specification applies to the components of the remote management system, whose construction and functional requirements are defined in the UNI/TS 11291 series.

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PREMISE

The technical specification is reviewed every three years. Any Comments on UNI/TS 11290-1 must reach UNI by February 2015.

This technical specification has been developed under the competence of of the body federated to UNI

CIG - Italian Gas Committee

The UNI Central Technical Commission has given its opinion approved on January 28, 2013.

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INDEX

INTRODUCTION 1

1 PURPOSE AND SCOPE 1

2 REGULATORY REFERENCES 2

3 TERMS AND DEFINITIONS, SYMBOLS AND ABBREVIATIONS 2

3.1 Terms and definitions 2

3.2 Symbols and abbreviations 2

4 CRITICAL ASPECTS OF OPERATIONAL SCENARIOS 2

4.1 General information 2

4.2 Opening/Closing the Gas Valve 3

4.3 Battery power supply 3

4.4 Communication time management 4

4.5 Access to the Wireless Communication Medium 4

4.6 Non-privacy inherent in wireless communication 4

4.7 PM1 communication subsections 4

4.8 Portable terminals and keys 4

4.9 Protocol and battery efficiency 4

4.10 Transfer of concession between distributors 5

4.11 Identifying vulnerabilities during the design phase 5

4.12 Critical aspects of the concentrator 5

5 PHYSICAL AND FUNCTIONAL PROTECTION OF EQUIPMENT 5

5.1 General information 5

5.2 Security Levels 5

5.3 Device Identifier 6

6 PHYSICAL AND FUNCTIONAL PROTECTION OF GdMs 6

6.1 Security levels for GdMs 6

prospectus 1

6.2 Hardware Protection 6

6.3 Flow Shut-Off Valve Management 7

6.4 Management and Protection of the GdM Battery 7

6.5 Periodic presence of transactions 8

6.6 Protecting the software installed in the GdM 8

7 PHYSICAL AND FUNCTIONAL SAFETY OF CONCENTRATORS 9

7.1 General considerations 9

7.2 Management and protection of the concentrator power failure state 9

7.3 Adoption of a secure operating system in the concentrator 9

7.4 Downloading the Application Software 10

7.5 First Installation/Configuration of the Concentrator 10

7.6 Storage and management of the keys of the GdMs present in the Concentrator 10

7.7 Synchronizing the Concentrator 10

8 PHYSICAL AND FUNCTIONAL PROTECTION OF TRANSLATORS AND REPEATERS 10

8.1 General information 10

8.2 Hardware Protection 11

8.3 Power Source Management and Protection 11

8.4 Software Protection 12

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9 **SAFETY RECOMMENDATIONS FOR THE SAC** 13

10 **PROTECTIONS IN THE APPLICATION PROTOCOL** General 14

10.1 information 14

10.2 Management organizational model 14

10.3 Category of A-PDU..... 15

10.4 Protection and structure of A-PDUs 16

prospectus 2 A-PDU Fields 16

10.5 Encryption and authentication algorithms and methods for non-copyability of A-PDUs 16

10.6 Encryption and Authentication Keys 17

10.7 A-PDU Numbering Template 20

prospectus 3 Example of response criteria and procedures..... 23

10.8 Specifics of Translator Encryption Keys 23

10.9 Operating states 25

11 **PROCEDURE FOR DOWNLOADING THE GdM SOFTWARE** 26

11.1 General information 26

11.2 Download procedure for GdMs communicating in PM1..... 26

11.3 Download procedure for GdMs communicating in PP4..... 27

BIBLIOGRAPHY 28

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INTRODUCTION

The Authority for Electricity and Gas (AEEG) issued, on 22/10/2008, the Resolution ARG/gas 155/08 “Guidelines for the commissioning of gas measuring groups, characterised by minimum functional requirements and with remote reading or remote management functions, for the points of redelivery of natural gas distribution networks”; amendments to the text of this technical specifications may become necessary in the future as a result of the measures that the AEEG should adopt within the scope of implementing the aforementioned resolution.

In general, the concept of "security" arises from the need to identify the set of appropriate and targeted measures for the "protection of an asset". In the scenario that follows will describe, the "good" to be protected is constituted by "information". Dealing with the "information security", according to consolidated and standardized practice, means identify protective measures to preserve:

- their "privacy" or "confidentiality" (counteracting the possibility that the content of the same may become known to persons not authorised to know it);
- their "integrity" (counteracting the possibility that they can be modified, tampered with, corrupted; in the worst case, ensuring that this can be detected);
- their "availability" (counteracting the possibility of loss, destruction, impossibility of access to them).

In a "remote management system", all peripheral devices (concentrators, translators, repeaters, GdM) belonging to it are to be considered as significant components of the system in the scenario that will be described.

Each of these components, in terms of “information security”, must be considered from two aspects:

- that of the device as a physical "container" of software and data to be protected;
- that of the apparatus as source/recipient in the exchange of the same in the communication with the Cent System.

In the case of the apparatus seen as a "container", it is necessary to foresee the use of appropriate physical, functional and installed software protection measures.

In the case of the device seen as "source/recipient", it is appropriate to provide adequate protection measures for the information exchanged, to be integrated into the protocols communication adopted.

In particular with regard to the protection measures to be applied in the protocols of communication, they should be:

- "encryption", in cases where "confidentiality" is required;
- the "authentication" of messages, which in addition to the "integrity" of the information exchanged also provides for the certification of their source;
- "non-copyability", to prevent the unauthorized cloning of exchanged information;
- the appropriate procedures and strategies for recovering information lost in the exchanges, to regain their "availability".

1

PURPOSE AND SCOPE

This technical specification defines the requirements relating to the safety of devices which are used in the remote management network and for the security of the data that is exchanged between the devices themselves.

This technical specification applies to the elements that make up the system of remote management, the construction and functional requirements of which are defined in the UNI/TS 11291 series.

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2 REGULATORY REFERENCES

This technical specification refers, through dated and undated references, to provisions contained in other publications. These regulatory references are cited at the appropriate points in the text. and are listed below. As regards dated references, subsequent amendments or Revisions made to such publications are valid only if introduced in this technical specification as an update or revision. For undated references, the latest version applies. edition of the publication referred to (including updates).

UNI/TS 11291 (all parts)	Gas measurement systems - Measuring devices of gas on an hourly basis
UNI CEI ISO/IEC 27001	Information Technology - Security Techniques - Systems Information Security Management - Requirements
ISO/IEC 27002	Information technology - Security techniques - Code of practice for information security management

3 TERMS AND DEFINITIONS, SYMBOLS AND ABBREVIATIONS

3.1 Terms and definitions

For the purposes of this technical specification, the terms and definitions set out in to UNI/TS 11291-1.

3.2 Symbols and abbreviations

For the purposes of this technical specification, the following symbols and abbreviations apply:

3.2.1 **A-PDU (Application Protocol Data Unit):** Protocol-related part of the message application.

3.2.2 **CSMA-CSA:** Acronym for Carrier Sense Multiple Access with Collision Avoidance, i.e. multiple access via carrier sensing that avoids collisions.

3.2.3 **DoS:** Acronym for "denial *of service.* " This refers to an attack computer science in which one tries to bring the functioning of a computer system that provides a service at the limit of its performance to the point of no longer being able to provide the service.

3.2.4 **HAN:** English acronym for "home area network" or communication networks between devices digital devices located in a "domestic" type area.

3.2.5 **LAN:** English acronym for "local area network" or communication networks between devices digital devices located in a limited area.

3.2.6 **WAN:** English acronym for "wide area network" or communication networks between devices digital sites located in an area that extends to one or more regions.

3.2.7 **PKI:** English acronym for Public Key Infrastructure.

4 CRITICAL ASPECTS OF OPERATIONAL SCENARIOS

4.1 General information

The operational scenario involving the GdMs communicating with the SAC via networks point-to-multipoint, is without doubt, in terms of "security", one of the most complex and difficult to address, due to the quantity and quality of potential points of vulnerability, problems and critical issues in it present, also due to functional, operational and organizational needs.

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For many of these GdMs, following the installation phase (which may also be "on the road"), it is standard practice not to perform routine periodic checks on their integrity (both hardware and software) and that of the system throughout their entire "operational life." They are physically inspected during their metrological verification, when they are replaced at the end of their "service life," when they are replaced due to a fault, or for routine operations such as battery replacement, when required. Considering that a battery's "useful life" can be up to seven years and that data reading, using wireless technology, does not require "visual" access, it follows that the GdM can remain unattended for significant, and in any case significant, periods of time. Under these conditions, the protection of each GdM against direct access to the hardware (in an attempt to tamper with the measurement system or the data, or to gain possession of the cryptographic keys) is entrusted to the intrinsic capabilities of the system to deal with such unauthorised direct access, which could also occur at a time prior to the installation of the GdM.

The requirement to use, for example and where possible, different keys for each GdM is part of the objective of limiting fraudulent actions, such as "key theft," to the targeted GdM. The operational scenario involving GdMs communicating with the SAC via point-to-point networks is, compared to the previous scenario, mitigated by the fact that they are generally more frequently accessed for battery replacement and/or routine periodic checks. While it should be noted that GdMs operating in this context, especially if they use public networks, could be subject to attacks by hackers who, primarily interested in disrupting service, could conduct such actions conveniently and regardless of proximity to the metering unit, communication between the GdM and the SAC is typically initiated by the GdM and generally continues for very limited periods of time. Furthermore, the transmission medium can be further protected directly by the operator against intrusion or interference by unauthorized parties.

The operational scenario involving Concentrators that communicate with the SAC via point-to-point networks is only partially mitigated by the fact that these devices are not directly involved in gas measurement and may be periodically visited for routine maintenance (e.g., battery replacement). On the other hand, the scenario is aggravated by the fact that they may contain the keys to hundreds or thousands of GdMs, which could become attractive targets for terrorist or fraud-related activities.

The operational scenario involving Translators and Repeaters must take into account that both the Translator and the Repeater do not maintain copies of the encryption keys of the "affiliated" GdMs and therefore the physical protection of said devices may be mitigated while in any case precautions must be taken to contain or mitigate DoS attacks.

4.2 Opening/Closing the Gas Valve

For GdMs that include one, the gas shut-off valve, which can also be operated remotely, is a particularly critical element. Control of this device is crucial for those operating the GdM, as losing control could also impact safety. Therefore, both the "physical" robustness of the GdM and the robustness of the communication protocols are equally crucial and fundamental in terms of safety.

4.3 Battery power supply

The presence of a power battery, which can be replaced if necessary, constitutes a potential vulnerability (for example, its theft would have the effect of a DoS attack, in addition to the economic damage caused by its restoration and the failure to collect measurement data, if no backup solutions are provided). This vulnerability should also be taken into account in the communication and operating modes of the devices, and in particular of the GdMs.

Furthermore, adequate disposal procedures for this component must be defined in a plan, agreed upon between the operators, with a view to protecting the environment.

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4.4 **Managing Communication Times:** The

goal of minimizing battery consumption requires adopting a communication strategy of alternating short "wake" periods with long "sleep" periods. This situation must be optimally managed in communication procedures.

For example, depending on the technology chosen, with different ratios between awake and asleep times, two scenarios arise: technologies with long sleep times (greater than 99.9% of the total time) are difficult to reach, while technologies that are virtually always awake can promptly report any identified attacks. It is necessary to optimally manage these situations in communication procedures to best contain/mitigate DoS attacks.

4.5 **Access to the wireless communication medium**

Wireless technologies intended for point-to-multipoint communications (where the "point" is the concentrator) have CSMA-CA access criteria. CSMA-CA channel access mechanisms are not immune to the problem known in the literature as "hidden terminal." This phenomenon occurs when two nodes are within range of a third node, but are not in direct visibility of each other. This increases the risk of collisions between transmitters, degrading communication quality and leading to transmission stalls. Consequently, communication procedures must also take this problem into account to prevent DoS attacks.

4.6 **Non-privacy inherent in wireless communication**

Wireless communication allows anyone to easily and anonymously spy on traffic and carry out software attacks (the most common being brute force and DoS attacks) within the wireless network's coverage range.

Note This range can vary from a few meters (LAN) to hundreds of meters (WAN) depending on the technology. used.

4.7 **PM1 communication sub-sections**

In PM1 communications, the communication link between the Concentrator and the GdM may include sub-links using various types of communication media and protocols. This could increase the hostility of the "communications network," especially when the technology used in these sub-links has a lower level of security than the chosen reference technology.

4.8 **Portable Terminals and Keys:**

When communicating locally with portable terminals for installation, maintenance, and operation (especially in the case of remotely accessible GdMs), the issue of protecting any cryptographic keys on the terminal itself must be considered, especially if it is owned by the company providing the local installation, maintenance, and/or operation services. This issue also needs to consider the possibility of theft/loss of portable terminals.

On the other hand, it is necessary to take into account the need to streamline the installation of the equipment in the field as much as possible, particularly during the mass replacement phase, combining this with the safety performance required by the protocol.

4.9 **Protocol Efficiency and Battery Life**

Communication protocols, particularly the application protocol, must also be designed with the goal of maximally conserving the battery life of system components, especially the GdMs (also to address targeted attacks such as DoS). It is therefore important to also consider the "efficiency" of the protocols so as not to underestimate their potential "verbosity."

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4.10 **Transfer of concession between distributors**

Taking into account that a remote management system may have multiple operators over time due to the of any "concession passages":

- the problem of certain and unequivocal identification must be taken into consideration of each component of the system to avoid confusion, in said steps of concession, both in the field and in the databases of remote management systems, especially as regards their identification as "devices" of the network of communication in routing management;
- the issue of integrity, availability and secrecy of the cryptographic keys of each component of the system, since these concession transfers take place between operators who, as provided for by the law, may find themselves competing to obtain them.

4.11 **Identifying vulnerabilities during the design phase**

When designing the system (communication protocols included), it is necessary to put maximum effort and commitment into identifying vulnerabilities and measures to be applied for their removal/mitigation (without obviously neglecting the issue cost-benefit analysis), since any vulnerability found subsequently upon commissioning, it may not be easy to contain it afterwards.

4.12 **Critical aspects of the concentrator**

Since the concentrator is a key element in the "point-to-multipoint" network of communication between the underlying SAC and GdM is essential, even for this apparatus, define all the significant topics that influence the implementation choices in terms of security. Vulnerabilities that may be present in the Concentrator may make some significant security measures applied on GdMs are ineffective and useless.

5 PHYSICAL AND FUNCTIONAL PROTECTION OF EQUIPMENT

5.1 General information

In accordance with fundamental security concepts, the vulnerabilities described above can be effectively countered by adopting the following basic criteria:

- security necessarily also requires the physical security of the device;
- Backdoors1) are not permitted in the production of the products, both hardware and software, useful for overcoming security barriers in case of recovery of the functionality of the devices. In fact, a backdoor, once discovered, can become an easy point of attack and functional tampering;
- no portion of the software resident in the devices may be based on other unsafe software;
- Manufacturers and distributors must have a safety-oriented organization that is able to manage, store, modify and distribute the parameters of security of all system devices by defining the roles and responsibilities of the actors of the organization and verifying with periodic checks that all the connected processes safety are compliant with the initial requirements.

5.2 Security levels

The requirements that the components of the remote management system or their parts to which they refer security features must satisfy, they are distinguished according to the levels2).

1) Backdoors in IT are comparable to "service doors" that allow you to partially or completely bypass the security procedures. These "gates" can be intentionally created by the manufacturers of the devices to allow for easier maintenance.

2) The levels are defined in ISO IEC 19790 and NIST-FIPS PUB 140-2.

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5.3 Device Identifier

To enable orderly management of the system also in relation to the change of concession, each device of the remote management system (GdM, Concentrator, Repeater, Translator), must have its own unique ID identifier using appropriate coding3). The ID, of appropriate size, must be printed in the factory in a manner permanent and unchangeable and must be visible even when the device is turned off.

For devices that communicate in PM1 the ID can also be used as device identifier for communication functions.

If the GdM is made up of multiple functional modules, each module must be equipped with its own ID but only the one relating to the device that manages the security must be valid as the device ID of the remote management/communication system.

6 PHYSICAL AND FUNCTIONAL PROTECTION OF GdM

6.1 Security levels for GdM

With reference to the safety requirements and in compliance with what is indicated in point 5.2, for the GdMs used in the remote management network created according to the requirements of the UNI/TS 11291 series, the following levels must be applied (see table 1).

prospectus 1

Security levels for GdM

GdM Class	Security level for PP4 communicating devices	Security level for communicating devices in PM1
B,C	2a)	-
A2, A1, A	2a)	2 a)

a) With single user authentication.

6.2 Hardware Protection

Protection of each GdM against direct access/attacks to hardware components (attacks aimed primarily at promoting fraud through tampering), namely:

- to the electronic boards and their components including the processing unit, memories, logic/analog gates, radio card, gas measurement sensors, battery,
- to the modules of which the GdM can be made up (see UNI/TS 11291-1),
- to any wiring present both between the components and between any modules of which above;

must be made with characteristics such as:

- in any case, provide for the closure of the above-mentioned components with containers mechanical solutions whose removal, in order to access the above mentioned components, can only occur with the breaking of elements of such solutions in so as to leave clear signs of any attacks;
- make any (non-destructive) attack on already installed GdMs impossible;
- deter potential attacks by making their activity on GdM difficult and prolonged not installed;
- have solutions in place that allow the GdM to highlight, record (on media (indelible) and communicate tampering and DoS attempts with particular reference to the wiring between components and/or modules;
- authenticate physical connections (using integrity control mechanisms) among the possible modules that constitute the GdM;
- ensure the inspectability of the system and physical connections, in compliance with the relevant standards applicable in the plant engineering sector;
- ensure the separation of the connections relating to the GdM from the plant those relating to other systems.

3) These codes are provided by the relevant bodies (for example the Flag Association).

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6.3 Flow shut-off valve management

The management of the valve used for the interception of the gas flow, in the GdM for the which function is intended, must meet the following requirements:

- the closing of the valve which determines the interruption of the gas flow must be able to be controlled locally and remotely only by means of appropriate commands authenticated and encrypted;
- the opening of the valve, previously closed, which determines the restoration of the operating condition in which gas supply can occur, it can be carried out only on site and only with prior consent to reopening given, by the site or remotely, by an authorized body.

6.4 Management and protection of the GdM battery

6.4.1 "Critical point" of the battery

Given that UNI/TS 11291 establishes that the "battery alarm" signal must occur at 10% of the battery's remaining autonomy, the **"critical point"** is defined of the battery that value of residual autonomy below which the GdM is no longer in able to guarantee all or even only part of its functions, including those relating to functional and measurement protections. For this purpose, each manufacturer must identify/define the value of the said **"critical point"** for your GdM .

The quantification of the critical point must be such as to allow, once reached, that:

a) the valve used to close/open the gas flow, if present in the GdM, block the flow; the

b) GdM maintains some or all of its functionality in relation to what declared by the manufacturer and within the constraints imposed by legal metrology.

The flow interruption feature when the battery reaches the critical point
It must also be possible to enable/disable it remotely.

6.4.2 Replaceable battery

In case the GdM is equipped with a replaceable power battery4), the GdM must provide placing a seal on the battery compartment cover; to prevent tampering even when storing the product, this seal must be affixed in the factory and downstream of each battery replacement. The GdM must be able to record in the event log the Battery replacement. In addition, at least class A1 and A2 GdMs must be equipped with a totalizer (which does not reset during the expected life of the device) that enumerates the seconds of operation from the moment of first ignition.

In order to deter battery theft, the battery should not be of the type used in the "electronic consumer" market, alternatively the battery must have characteristics exclusive physical/geometric.

Battery replacement should be performed by authorized personnel and subject to authorization from the GdM itself which, for this purpose, should be previously prepared for replacement.

Following and during the replacement of the battery, the valve in the GdMs for which it is intended, must remain in the position it was in before the replacement.

If the battery replacement has not been previously authorized by the GdM or the valve is extended beyond a configurable time (tsb) (for example tsb = 2 min) in the GdM for which it is intended, it must be placed in the CLOSED state and stop the supply of the gas. This function must also be able to be deactivated/activated remotely. The opening of the valve previously closed due to lack of authorization to battery replacement or prolonged absence of power supply must be able to be carried out by authorized personnel.

4) "Replaceable battery" means that the replacement of the battery, due to normal wear and tear or decay of the its capacity, does not produce the removal of metric seals.

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When at least the functional measuring module of the GdM is in the state following the unauthorized removal of the battery or failure of the battery, it must ensure and keep active in this state, for a time not less than td (power-off time), the measurement of volumes (at least at the thermodynamic operating conditions - Vm). The td value must be indicated by the manufacturer and in any case not less than 70 d in the declared conditions.

6.4.3 Built-in non-replaceable battery

In case the GdM is equipped with one or more non-replaceable batteries, the GdM must be prepared with one or more totalizers (which do not reset during the expected life of the device) that lists the seconds of operation from the moment of first ignition.

In order to deter battery theft, the battery should not be of the type used in the "electronic consumer" market; alternatively the battery must have characteristics exclusive physical/geometric.

6.4.4 End-user replaceable battery

In case the battery is allowed to be replaceable by the end user, its physical characteristics may also be equal to the commercial ones, differently as indicated in 6.4.2.

6.5 Periodic presence of transactions

In order to deter DoS attacks (exercised for fraud purposes), the GdMs that implement the valve for interrupting the service, following its first acceptance by the network communication, must list the days of continuous absence of communication with the Concentrator/SAC (IUT). If it happens that IUT > IUT_max (with IUT_max (also configurable remotely) the GdM stops the supply by closing of the valve. This function must also be able to be deactivated remotely.

6.6 Protecting the software installed in the GdM

The following are measures to enable the detection of any tampering and/or modifications and/or deliberate corruption of the software installed in the GdM.

In order to allow the verification of the "genuineness" of the software installed in the GdM of class A2 or A1, said software must be accompanied by a "Certificate of authenticity". This certificate may consist of a file type document (for example in text format) containing at least:

- the manufacturer's personal details;
- the version data of the software whose authenticity is certified;
- any release notes;
- the "fingerprint" of the software to be certified, calculated with an algorithm trusted (such as MD5 or SHA-1 or SHA-2 or others).

This file must be signed using a digital certificate owned by the manufacturer (which thus cannot repudiate the software). These provisions must apply to both the initial version of the software for subsequent updates. It is strongly recommended

that this certificate be installed in the GdM (in an appropriate part dedicated non-volatile memory of the GdM), and updated simultaneously with the "download" of the software to which the certificate is associated. The "verification" of the installed software with the relevant "Certificate"

"of authenticity" can be carried out offline, using appropriate equipment.

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7 PHYSICAL AND FUNCTIONAL SAFETY OF CONCENTRATORS

7.1 General considerations

The concentrator is the element of the gas remote management system capable of safely managing the underlying GdMs, safeguarding and protecting access to the data stored within it.

The concentrator plays a key role in system security, storing a copy of the data and managing access to the underlying network elements and the SAC. The concentrator must protect the data from unauthorized access through the various local and remote ports and must only allow access to authorized entities.

The concentrator must allow connections only through identified ports and protocols: insecure remote connections (for example: Telnet:23, HTTP:80) are expressly prohibited.
If multiple operator profiles can access the Concentrator, the latter must manage at least three different operator profiles (Administrator, Remote Operator, Local Operator), each with their own credentials and authorizations.

For communication with the SAC, the concentrator must provide a secure channel according to pre-established security profiles, using tools generally available on public communication networks.

The concentrator is not considered a "trusted device" and as such cannot autonomously modify the security parameters of the underlying devices.

Every time a new meter is enrolled in the application, the concentrator receives all the associated security parameters from the SAC. These parameters must also be stored securely on the concentrator.

The Concentrator must be designed so that the guaranteed security level is at least level 35).

7.2 Management and protection of the concentrator power failure state

The concentrator is a device generally powered by primary sources such as the electrical grid, solar panel or other power sources with high availability and is equipped with an appropriate auxiliary power source, as prescribed in UNI/TS 11291-7.

The connection to the primary power source must be made in such a way that it is easily inspected and does not allow accidental interruption of the power supply. Any electrical safety and disconnect devices placed between the concentrator and the power source must have the same degree of protection as the concentrator and must be designed to easily detect tampering or intervention by unauthorized personnel.

During power outages from the primary source, it is necessary to implement measures such that switching to operation using reserve sources does not impact the safety of the concentrator.

While the concentrator is powered only by the secondary source, communication to the local terminal and all anti-fraud functions, with subsequent communication to the SAC, must be maintained active for a time (tdr) declared by the manufacturer, in any case no less than 30 days. Once the tdr time has elapsed , all functions may be interrupted, but in any case, it must not be possible to access the security parameters (e.g., encryption keys) and sensitive data stored in the concentrator (e.g., encrypted data received from the GdMs).

7.3 Adoption of a secure Operating System in the concentrator

If the concentrator software architecture requires the adoption of an Operating System (whether custom or purchased), this must implement well-documented primitives and functions for security management.

5) The levels are defined in NIST FIPS PUB 140-2.

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7.4 Downloading the Application Software

Downloading the Application software must be possible either via local port or via WAN interface.

Updating the software must be done using objects and procedures defined in the PP3 protocol (see UNI/TS 11291-8).

The new software, in order to guarantee its correctness, authenticity and genuineness, must be digitally signed using certificates in the PKI environment.

7.5 First Installation/Configuration of the Concentrator

The first installation, as for the GdMs, must be carried out in absolute safety for example by adopting procedures similar to those described for the GdMs (opening of session local, followed by a self-consistent secure message that establishes a session key temporary on the basis of which authenticated data exchanges take place).

Subsequent accesses to the concentrator for maintenance purposes must follow at least the rules of the following safety measures:

- access by the operator identified on the basis of security credentials;
- initial temporary session key establishment message;
- exchange of at least authenticated data;
- session closure with destruction of the temporary key.

7.6 Storage and management of the keys of the GdMs present in the Concentrator

The keys of the GdMs underlying the concentrator are transferred from the SAC to the concentrator every time a GdM is recruited or whenever for some reason the keys are updated.

The following describes an example of implementation of archiving and management of keys that meet the security requirement imposed by level 36).

The keys are stored in a database encrypted with a symmetric key, provided by the SAC to the concentrator in a secure manner and this database resides in a security module dedicated and protected.

The use of the specific key in a data exchange between Concentrator and GdM should occur with the mediation of this module which receives the data block to be authenticated and/or encrypt/decrypt and the GdM identifier returns the data block encrypted/decrypted or the MAC field without any external exposure of the keys.

In case an attempt of unauthorized access or tampering is detected, database, should at least be immediately deleted (securely) database encryption key.

7.7 Concentrator Synchronization

The Concentrator can be synchronized by the SAC or adopt self-management procedures synchronization as long as they are deemed safe.

In case it is not possible to proceed with clock synchronization for a TAS time (synchronization absence time) configurable from SAC, the Concentrator must suspend synchronization activities on the devices affiliated with him.

8 PHYSICAL AND FUNCTIONAL PROTECTION OF TRANSLATORS AND REPEATERS

8.1 General information

In the context of the remote management network defined in UNI/TS 11291-1, the Translators are remote management system devices capable of carrying out a translation of at least one of the first three ISO-OSI layers (physical; data link; transport) of the communication protocol between GdM e Concentratore o SAC.

6) The levels are defined in NIST FIPS PUB 140-2.

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In the context of the remote management network defined in UNI/TS 11291-1, the Repeaters are elements network auxiliaries capable of extending PM1 communication radio coverage; they operate only on the "data link" and "network" protocol levels, they process information necessary for communication but do not manage GdM data at the application level.

In the context of the following points the GdMs or Repeaters associated and managed by a Translator or by the Repeaters are referred to as “affiliate” devices.

The task of Repeaters and Translators is to retransmit correctly received messages. by their affiliates without entering into the context of the data contained in the message and intended for the application level of the SAC or Concentrator; Repeaters and Translators not They therefore need to manage their affiliates' encryption keys.

With reference to the critical aspects of the operational scenarios discussed in point 4 and in compliance as indicated in point 5.2, to the Translators and Repeaters used in the remote management network, made according to the requirements of the relevant parts of UNI/TS 11291, can be Security level 1 applied.

As regards the management of the specific functions of the translators and repeaters, as well as for the Authentication is required to configure them, whether remotely or via a local interface.

8.2 Hardware Protection

With reference to the minimum level of security that must be guaranteed by the Translator/Repeater It is necessary that the Translator/Repeater makes attempts to intrusion or tampering aimed at:

- to the electronic boards and their respective components, including the control unit processing, memories, logic/analog gates, radio card, battery;
- to the modules that the Translator/Repeater can consist of;
- to any wiring present both between the components and between any modules above.

Therefore the Translator/Repeater hardware must be built in such a way as to:

- in any case, provide for the closure of the above-mentioned components with suitable lids/caps/containers which must be removed to access the above mentioned components, can only occur with the breaking of said lids/caps/containers or after removing mechanical seals in such a way to leave clear signs of any attacks;
- as an alternative to the above point, have solutions such that the Translator can highlight, record (on indelible media) and communicate attempts to tampering and DoS with particular reference to power supply and wiring between components and/or modules;
- authenticate physical connections (using integrity control mechanisms) between any modules that make up the Translator/Repeater;
- ensure the inspectability of the system and physical connections;
- ensure the separation of the systems (of the physical connections relating to the Translator/Repeater compared to those relating to other systems).

8.3 Power source management and protection

8.3.1 Battery critical point

For Translators/Repeaters powered only by battery, the "critical point" of the battery that value of residual autonomy below which the translator is no longer in able to guarantee all or even only part of its functions, including those relating to functional protections. For this purpose, each manufacturer must identify/define for your Translator/Repeater the value of the said "critical point".

When the battery reaches its “critical point" the Translator must stop all or some of its features in relation to what is declared by the manufacturer and in any case case the function of synchronizing the clocks of the "affiliates".

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When the battery reaches its "critical point" the Repeater must cut off all or some of its features in relation to what the manufacturer declared.

8.3.2 Power case with replaceable battery

In case the Translator/Repeater is equipped with a replaceable power battery, it must provide for the application of a “seal” on the battery compartment cover; for highlight attempts at tampering even during product storage, such The seal must be affixed at the factory and after each battery replacement.

In order to deter battery theft, the battery must not be of the type used in "electronic consumer" market; alternatively the battery must have characteristics exclusive physical/geometric.

Battery replacement must be done with the prior authorization of the Translator/Repeater otherwise an event must be generated.

8.3.3 Case of power supply from another source

In case the Translator/Repeater is powered by the mains or by a solar panel or from another source external to the Translator itself, it is necessary that the connection to the network electric or solar panel or external source is made without any solution continuity while also ensuring that no device can interrupt the power supply is accessible without causing permanent breakage of special seals physical that can only be restored by authorized personnel.

8.3.4 Management of the "power off" state

When the Translator/Repeater is in the power-off state following the removal of the primary power source (battery, mains, panel solar, etc.) it must guarantee and maintain active in this state, for a period of time not less than td, at least features such as time counters, event log, those any and all relating to the detection of tampering. The time td must be declared. by the manufacturer and in any case not less than the value indicated in UNI/TS 11291-7.

8.4 Software Protection

The following information is intended to indicate measures to enable the detection of any tampering and/or modifications and/or voluntary corruption of the installed software in the Translator/Repeater.

In order to allow the "genuineness" of the software installed in the Translator/Repeater said software must be accompanied by a "Certificate of authenticity". This certificate can be made up of a file type document (for example in text format) containing at least:

- the manufacturer's personal details;
- the version data of the software whose authenticity is certified;
- any release notes;
- the "fingerprint" of the software to be certified, calculated with an algorithm trusted (such as MD5 or SHA-1 or SHA-2 or others).

This file must be signed using a digital certificate whose manufacturer is the owner (who thus cannot repudiate the software). These provisions must apply both for the initial version of the software and for subsequent updates. It is strongly recommended that this certificate be installed in the Translator/Repeater (in appropriate dedicated part of non-volatile memory), and updated at the same time as the "download" of the software to which the certificate is associated. Verification of the "authenticity" of the installed software can be carried out by anyone, authorised, who requires such verification.

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9 SAFETY RECOMMENDATIONS FOR THE SAC

The SAC is not part of the distribution network, however being in communication with components of such a network equipped with keys whose secrecy is fundamental for security and receiving from such components information whose confidentiality must be protected, the SAC also must meet safety requirements without which it would itself constitute the weak link in the security chain, thus risking becoming the "back door" for the entire system.

Unlike other devices in the system, the SAC cannot be seen as a single apparatus with a well-defined physical boundary, nor as a set of apparatuses with such a feature. On the contrary, the SAC must be seen as a logical component that can be distributed on physical nodes on which other components that do not do the same also reside part of the SAC itself.

The general safety requirements of the SAC are defined in UNI CEI ISO/IEC 27001 and ISO/IEC 27002 and must comply with the requirements of the legislation in force7).

The general safety requirements that all the devices must meet as a whole possible components of the SAC are in summary8):

- **Access Control:** The Service Manager must limit access to the SAC to authorized users, automated processes, functions or transactions acting in the name and role of authorized users, to devices or other accredited information systems.
- **Identification and Authentication:** The Service Manager must identify and authenticate SAC users, automated processes, functions or transactions acting in SAC comparisons in the name and role of authorized users, devices and other systems information that interacts with the SAC.
- **Physical and environmental protection:** the Service Manager must:

  i) limit physical access to the systems and operating environments on which the SAC is installed by authorized individuals;

  ii) protect the facilities and infrastructures on which the SAC bases its operations functionality;

  iii) ensure the availability of resources that maintain service continuity from the SAC;

  iv) protect the SAC from environmental risks;

  v) provide adequate environmental controls in the facilities hosting the servers on which the SAC components are installed.
- **Protection of Systems and Communications:** The Service Manager must:

  i) control, manage and protect communications within the SAC perimeter; ii) adopt, design and develop systems and methodologies that guarantee the efficiency of communications, particularly between the SAC and field devices.
- **System and Information Integrity:** The Service Manager shall: i) identify, report and correct system and information defects managed in a timely manner;

  ii) provide adequate protection from malicious code ("malware") at appropriate points within the information systems that host the SAC.

The main safety concern in the SAC is the management of the keys to the field equipment. In this regard, it is specified that the responsibility of the SAC can be restricted to the *"run time"* management of the keys and not necessarily extended to their safe storage for an indefinite period. The Service Manager is required to securely store keys indefinitely, but this feature is not necessarily internal to the SAC, if the Service Manager has adequate reliable system for storing authentication secrets, possibly shared in use also by other systems.

7) On the date of publication of this technical specification, Legislative Decree 30 June 2003, No. 196 is in force. "Personal Data Protection Code".

8) See NIST FIPS-200.

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In the event that a secure secret repository external to the SAC is not available, the secure storage of field device secrets becomes the responsibility of the SAC.
In this case, at least level 39 must be guaranteed (for example, the SAC could store the secrets in encrypted form and the encryption key used to perform this encryption must not reside in the same component or database that stores the encrypted secrets).

Another important functional responsibility of the SAC is to ensure accurate and reliable control of the synchronization of the devices with which the SAC is in direct communication (concentrators, translators, GdMs in point-to-point communication).
To ensure this requirement with a sufficient level of security, the Service Provider must identify reliable sources of synchronization services (e.g., NTP servers) and provide the SAC with a connection to at least two of these sources. Based on this redundancy, the SAC must verify that the current time values from all sources are equivalent and use this value to synchronize its local clock. In the event of a discrepancy between the synchronization sources or a connection failure with one of the sources, the SAC must notify the Service Provider and suspend the synchronization activity of the devices enrolled in it.

10 PROTECTIONS IN THE APPLICATION PROTOCOL

10.1 General information

The ISO-OSI architectural model, structured across seven layers, is used as a reference for application protocol frames and procedures. In this model, application protocol frames (called "application messages") are identified by the acronym A-PDU (Application Protocol Data Unit). This acronym is used below to unambiguously distinguish these types of frames. Each A-PDU is exchanged "encapsulated" in the protocols used in the various communication links and defined in UNI/TS11291-8.

In the following, "Operator" is the entity that integrates the SAC components and, if applicable, but not necessarily, concentrators or translators used for communication with the GdMs, except in cases where the distinction between said components is explicitly stated.

10.2 Management organizational model

In the remote management system compliant with the requirements of AEEG resolution ARG/gas 155/08, an organizational/management model is identified and defined in which the following "actors" can operate, each with their own distinct roles and functions:

- *Operator:* the person who manages the fleet of GdMs and other system components. This role may be assumed entirely or partially directly by the distributor who obtained the GdMs under concession, or entirely or partially by the entity offering/providing the operation services to the distributor. In the latter case, the distributor must provide the Operator with all the information necessary to operate the GdMs and/or other system components under concession.

- *Guarantee Authority:* A guarantee institution that must provide, at the request of the incoming Operator, the assistance service for the reconfiguration of the GdM's "operating" cryptographic keys during the transfer of their concession; or alternatively, in cases where the Operator has lost said keys during normal operation. This is done with the aim of ensuring their confidentiality and exclusive access to the Operator operating the GdM fleet. The *Guarantee* Authority is the sole holder of the service keys. The role of Guarantee Authority may be assumed by the Operator or the Manufacturer of the equipment, or by a legally recognized third-party entity, in the event that this guarantor role is not established. In any case, it is recommended that the incoming Operator change the encryption and authentication keys of the devices received in the change of concession as quickly as possible to mitigate any risks of attacks such as theft (interception) and/or improper modification of the keys.

9) The levels are defined in NIST FIPS PUB 140-2.

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- *User:* the end user/consumer of the gas distribution service, with the associated GdM for measuring consumption.
- *Installer/ Maintenance Technician:* person responsible for the installation and/or maintenance of GdMs.

10.3 **A-PDU Categories** The

"actors" described above consequently lead to the definition of the following four "end-to-end" A-PDU categories, each, as a rule, associated with the function performed by each "actor".

- "Service" A-PDUs , used by the Guarantee Authority or its representative to reconfigure the keys when the concession is transferred;
- "Operating" A-PDUs , used only by the Operator in the normal operation of the remote management system;
- "User" A-PDUs , used, when required, for the exchange of information with the User (for example for "home display" applications);
- "Installation/maintenance" A-PDUs , used by the maintenance/installer and the Operator for maintenance of the remote management system.

By "end-to-end" A-PDU we mean:

- A-PDUs whose structure and content remain unchanged (by virtue of the "encapsulation" process already described) in the exchange of the same between one interested party and another, positioned at one extreme and the other of the exchange;
- A-PDUs authenticated and (where applicable) encrypted, during sending, with shared keys known only to the parties involved in the exchange of said A-PDUs, so that only those parties, during the receiving phase, will be able to verify their authenticity and (if encrypted) read their contents (see details in the following points); - A-PDUs containing information for identifying the sender and for the correct reading/

interpretation of the A-PDU by the recipient.

The "Initiator-Responder" communication scheme is used for the application exchange procedures of A-PDUs. The "Initiator" controls the exchange procedures, since it initiates the transaction on its own instance with a request A-PDU; the "Responder" sends the resulting response A-PDU to the instance received from the "Initiator".

In this “Initiator-Responder” communication scheme, the distinction between:

- "Inline" A-PDUs, those that "Initiator" and "Responder" exchange directly for example: - between the Merchant (Initiator) and the GdM (Responder); - between the portable terminal (Initiator) and the GdM (Responder); - between the GdM (Initiator) and the Merchant (Responder); - between the GdM (Initiator) and the User's Device (Responder).
- "Offline" A-PDUs, those that the "Initiator" and "Responder" exchange indirectly, such as those that the Merchant (Initiator) and the GdM (Responder) exchange through the portable terminal (which acts as an "offline" repeater) or any other repeater devices. These A-PDUs, by virtue of the "encapsulation" process, transit through the various intermediate communication links in compliance with the "end-to-end" logic.

As for the Guarantee Authority, the adopted framework should take into account that it may not have its own communication system (SAC and respective concentrators) but simply an interface (e.g., web) provided by the Operator to allow the Authority to provide the assistance services requested by the Operator. In this context, it would remain the Operator's responsibility to identify the correct routings to allow the exchange of A-PDUs with the interested GdMs.

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The use of A-PDUs can be outlined as follows:

- for "service" A-PDUs, taking into account that the Guarantee Authority does not have to be required to know the routes applied by the Operator, will use A-PDUs of "service" for "offline" exchanges as the most conservative choice;
- for "operating" A-PDUs, their use must be foreseen for both "on-line" exchanges be "out of line";
- for "user" A-PDUs, their use must be foreseen only for "online" exchanges;
- for "installation/maintenance" A-PDUs, their use must be foreseen only for "online" exchanges;

It is important to highlight that the "service" A-PDUs and the "operation" A-PDUs of the "outside" type "line" allow you to avoid carrying keys on board portable terminals "service" or "operation" cryptographic features of the GdM, thus avoiding risks of compromise of the keys themselves.

10.4 Protection and structure of A-PDUs

Protection of A-PDUs must be achieved through "encryption", "authentication" and methods of "non-copyability" of the same. In addition to this, it must be added and specified that:

- an encryption and authentication model must be used that offers performance of protection no less than that which would be obtained by using "keys shared symmetric" of at least 128 bits10);
- Encryption must be applied in cases where the information transported will be among those classified as "sensitive/confidential" by law national/international and in cases of "service" A-PDU exchanges; it can be also applied to any other information that the Operator deems, at its discretion, judgment, appropriate to protect;
- A-PDUs, even if not encrypted, must always be authenticated and not copyable.

As regards the structure of the A-PDU, this must have characteristics such as minimize the redundancies of the A-PDUs themselves to optimize their efficiency. The general structure of the A-PDU should include the fields shown in Table 2 (not necessarily in the order illustrated).

prospectus 2 **A-PDU Fields**

Header	Body	Authentication code

- "Header" of the A-PDU, always transmitted in clear text; the information in it contained must serve for a correct interpretation of the entire A-PDU.
- "Body" of the A-PDU, containing the useful information to be transmitted, encrypted if deemed confidential or clear where confidentiality is not necessary.
- "Authentication Code" (CA) of the A-PDU, of size not less than 64 bits for the A_PDUs used in PM1 communications and 32 bits for A-PDUs used in point-to-point communications, also transmissible in clear and calculated form, with the algorithm of authentication chosen, on the information contained in the "header"11) and the "body" in clear or already encrypted A-PDU. This calculation can also include information not expressed in the "header" or in the "body" of the A-PDU but implicitly known and shared by "Initiator" and "Responder" (such as the GdM ID).

10) These specifications are contained in NIST publications.

11) Information possibly contained in the A-PDUs but relating to the operation of the layer may be an exception. transport.

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10.5 Encryption and authentication algorithms and methods for non-copyability of A-PDUs

The **"encryption"** and **"authentication"** of the A-PDUs must be accomplished using accredited and consolidated algorithms that use symmetric keys of at least 128 bits, such as the following:

- AES128-CMAC for authentication12);
- AES128-CTR for encryption13);
- AES 128 –CBC for cipher13);
- AES128-GCM for encryption and/or authentication14);
- AES128-GMAC for authentication14).

The AES-ECB (Electronic Code Book)13) algorithm cannot be used because it does not deemed sufficiently secure to ensure data confidentiality.

The protocol must also include mechanisms for the "non-copyability" of the A-PDU.

Only A-PDUs received (from the "Initiator" or from the "Responder") with a consistent authentication code (CA) and compliant with the rules established in the protocol regarding their "non-copyability".

10.6 Encryption and authentication keys

The following types of symmetric keys (each at least 128 bits) must be provided: encryption/authentication for each GdM, each associated with the category of the Application message to protect:

- "service" key;
- "exercise" key;
- "user" key;
- "installation/maintenance" key.

For the keys of each of these types, the requirements indicated in the following points.

10.6.1 General characteristics and key management

The encryption and authentication key of the same A-PDU can be identical15).

Encryption/authentication keys must never be transmitted in clear text.

The value to be assigned to each key must be generated randomly, unpredictable and complex16). Keys with all bits set to one are not allowed.

Keys with a null value (all bits zero) are by definition "deactivated", meaning that The "Responder" must ignore the messaging associated with them.

Encryption keys must be stored in the devices on non-volatile media, must be editable (overwritable), but not readable from the outside.

All key changes must be logged in the event log.

- Each A-PDU detection received with congruent numbering and incongruent CA (an event mainly associated with possible attempts to attack with non-keys correct), must be recorded (event log) or at least enumerated in a non-resettable register.

10.6.2 Initial key setup

- The "service" key, when used, is initially configured at the factory and released (together with the device ID) to the Guarantee Authority in a secure manner.

12) NIST view – SP 800-38B.

13) See NIST – SP 800-38A.

14) See NIST – SP 800-38D.

15) Notwithstanding the provisions of NIST SP800-57.

16) See ANSI X9.31 Appendix A.2.4.

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- The "operator" key is initially configured at the factory and released (along with the device ID) only to the distributor who purchased the device; if another party acts as the Operator, the distributor itself releases the same information to that party in a secure manner.

- The "user" key can be initially configured at the factory or by the operator during the operation phase. In both cases, the operator, or an entity authorized by the operator, always issues it to the user in a secure manner.

- The "installation/maintenance" key, for the initial installation phase of the GdM, if equipped with a local communication port, can be factory-configured to a default value (which can also be the same for all devices in the same category) and assigned a duration sufficient to ensure successful installation. For the initial installation of GdMs not equipped with a local communication port, the installation/maintenance key can be generated in pseudo-random mode (values with all bits set to "1" or "0" are not permitted) automatically by the GdM and communicated to the installer, without being transmitted via radio (for example, the GdM can display an 8-digit "seed" that the GdM and the maintenance terminal will use to calculate the key according to a shared procedure). For the maintenance phase of the GdM (following the installation phase), the installation/maintenance key is configured/activated via appropriate "operation" messages. The installation/maintenance key, especially during the initial installation phase, must be assigned limited privileges so as not to compromise the security of the device.

- The transfer of the operating and service keys from the Manufacturer to the Operator should take place by transferring the value of the new key using "Key Transport" methods17) (for example after having undergone a "key encapsulation" process which uses the public key of an Operator's certificate as the encapsulation key).

10.6.3

Key Properties The "service"

- key has an unlimited life; it can only reconfigure itself and the "exercise" key; it can only be reconfigured with appropriate "service" messages and only with a new key with a non-zero value; it must always be possible to restore the value of the key originally configured at the factory using appropriate service messages; it cannot be "deactivated" (therefore it is always active, i.e. non-zero).

- The "operation" key has an unlimited lifespan; it can reconfigure itself, the "user" key, and the "installation/maintenance" key; it can be deactivated and reconfigured with appropriate "service" or "operation" messages; it allows all scheduled operating activities to be performed.

- The "user" key has an unlimited lifespan; it can be deactivated; it can only reconfigure itself; it can be reconfigured with appropriate "operation" or "user" messages; it only allows the management of messages intended for the user.

- The "installation/maintenance" key has a limited lifespan; it cannot reconfigure itself but can be reconfigured with appropriate "service" messages; the activities that can be performed with it and its lifetime during installation are defined; it deactivates automatically upon expiration of its lifetime or artificially with appropriate "service" and/or "installation/maintenance" messages. The "installation/maintenance" key must be associated with "privileges" that identify which activities (functions) are permitted.

10.6.4 Key custody

The "service" key must remain stored only in the central system of the Guarantee Authority.

The "operation" key must remain stored in the Operator's central system and at most also in the concentrator (not the translator) if the latter has adequate security requirements to be considered an "appendage" of the central system.

17) In compliance with NIST SP800-57.

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For the "user" key, it is the User's responsibility to keep it safe after its release. by the Operator.

Regarding the "installation/maintenance" key, for the maintenance phase, it is responsibility of the maintainer for its custody, after its release by of the Operator.

10.6.5 **Specificity of class A1, A, B and C GdMs** For

class B and C GdMs (see UNI/TS 11291-1) and for class A1, A GdMs which communicate with the SAC in point-to-point mode, taking into account the characteristics of the vehicle transmission and what is required by UNI/TS 11291-4 and UNI/TS 11291-5, must be considering the additional provisions referred to in this chapter.

10.6.5.1 Profiles and login credentials

GdM with meter classes > G10 that communicate with the SAC in point-to-point mode, they are generally divided into simple devices (class A1 and A) with 1 access profile, and in advanced devices (Class B and C) that can manage up to five access profiles18).

A separate operating encryption key must be provided for each access profile. and the possibility of managing different operators, each of which must be able to be associated with a different login credential.

The access credential is used to identify the operator, in communications with the GdM.

One of the above profiles is Service Administrator (AS - service administrator), and every other profile must have its own administrator (AP – profile administrator) who enjoys superior privileges compared to other operators of the same profile.

Each AS operator has the following privileges (not modifiable):

- reading and writing all variables;
- enabling read and write privileges for each variable or function for the other profiles;
- ability to change the encryption keys of all profiles (including the one for the profile AS itself);
- ability to modify only the login credentials of the administrators of the remaining profiles (including your own).

Each AP operator has the following privileges (not modifiable):

- ability to change the encryption key of your profile (if the function has been enabled previously authorised by one of the AS operators);
- possibility to cancel/modify the access credentials of the operators own profile (including your own).

10.6.5.2 Requirements for Managing Access Credentials and Encryption Keys

In addition to the safety requirements expressed in the previous chapters, the class B and C GdMs and those Class A devices communicating in point mode must comply with the following requirements:

- identification login credentials must never be transmitted in clear text;
- Each profile is associated with an Exercise Key (Kc). The exercise key of each profile, other than AS, can be modified by the administrator of that profile, if enabled the privilege to write to that key from the AS, and from an administrator of the service (AS);
- The AS profile has all write and read privileges and cannot be inhibited; each AS profile operator has the ability to dynamically assign privileges to other profiles writing/reading each variable and executing each function;
- The temporary installation/maintenance key is associated with the AS profile only. Therefore the activation of the installation/maintenance key must be done with an appropriate command encrypted with the AS profile operating key;

18) For the definition of “access profile” see UNI/TS 11291-1.

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- the GdM, when the installation/maintenance key is active, must reject any encrypted command with the installation/maintenance key which involves the modification of keys and/or access credentials, any command that allows the deactivation of logical seals and commands aimed at updating the software;
- All communications with the SAC must always be authenticated; they can do so except those relating to service messages (such as confirmations and denials when executing commands or requesting identification);
- Key changes, of any profile, must be recorded in a specific register that cannot be erased except locally and after the removal of a specific tool seal;
- any attempt to communicate with the GdM with an incorrect encryption key must be recorded in the event log or enumerated in a non-resettable log;
- any attempt to access the GdM by an unidentified operator (with credentials of (erroneous access) must be recorded in the event log or enumerated in a register not resettable.

In the first installation phase the GdM activates the key default installation/maintenance configured at the factory for the AS profile: the installer use the installation/maintenance key for message security useful for the first configuration of the device. At the end of the installation phase the key installation/maintenance must be able to be made inactive.

Any of the AS profile operators can modify, using the operating key of the profile, the operating key of your profile and that of the remaining profiles when managed.

Any of the AS profile operators can modify/activate, using the key exercise, the installation/maintenance key.

At least the AS profile operating key should be changed periodically.

For GdMs that manage more than one profile (for example class B and C GdMs) one of the four available profiles can be reserved for the "maintenance service" whose operators The AS assigns the privileges necessary for complete and correct maintenance of the GdM: each maintainer authenticates with the GdM always with that profile using its relative operating key.

For GdMs who manage only one profile (the administrator one), the key installation/maintenance allows you to carry out maintenance on the equipment personnel who may not be granted knowledge of the key operation. For this purpose, the installation/maintenance key must be activated be carried out by means of an encrypted command with the operating key and must be able to occur without the maintenance personnel knowing the key of operation in use (for example with messages containing A-PDUs "off line").

10.7 A-PDU Numbering Template

10.7.1 General information

Without prejudice to alternative solutions, already defined by application protocols approved at the national level international, a structured and well-defined A-PDU numbering model must be provided organized in such a way as to possess the following requirements and provide the following services:

a) be able to guarantee, for the A-PDUs of each type, the uniqueness of those that potentially could be exchanged in a time frame no less than 15 years (for example 20 years as a conservative value);

b) allow that:

- the numbering of A-PDUs of each type can be independent and uncorrelated to the numbering of A-PDUs of other types for the purpose of eliminate any possible mutual conditioning between A-PDUs of the different sets, in the phases of their operational use,

- in the case of A-PDUs of the "Broadcast" or "Multicast" type, it may not be foreseen no numbering as the non-copyability of the message must be made with different application methods;

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c) allow, for A-PDUs of each type and where it is intended to be adopted, to put in place relate the number assigned to each A-PDU to the amount of A-PDUs actually exchanged, in order to allow:

- the reconstruction of the order/consecutiveness of the received A-PDUs, reconstruction essential in cases where the order/consecutiveness of the information contained in the A-PDUs received was necessary,
- the timely detection of the presence of any anomalies found in the numbering of the A-PDUs received, attributable, for example, to loss of A-PDUs, to malfunctions in the numbering process, to exchanges of false A-PDU,
- the use of automatic and timely methods for recovering lost A-PDUs above all efficient;

- the use of the "sliding window" acceptance criterion for the received A-PDUs from the "Responder", consisting in the acceptance of the A-PDUs deemed "valid" whose progressive numbering will be included in a window made up of limits predefined numbers; point 10.7.2 provides details on the characterization of the said "sliding window" and the definition of the meaning of "validity" of an A-PDU;

d) exclude that such model can be conditioned by synchronizations/alignments between the clocks of the interlocutors involved in the A-PDU exchange;

e) allow the resetting of the progressive numbering of the A-PDUs of the types belonging to a category, in conjunction with the configuration/reconfiguration of the encryption/authentication key associated with that category (this operation for example, it is particularly useful to the incoming Operator in cases of reconfiguration of the "operation" keys of the GdMs, by the Authority of Guarantee, during the concession transfer phases).

10.7.2 Acceptance criteria and procedures with a "sliding window" of the A-PDUs received from the "Responder"

The requirements of the “sliding window” and the meaning of it are described and defined below. "validity" of an A-PDU received from the "Responder".

For ease of reference, the following is reported:

- In the following, "congruent" A-PDU is used to indicate a received A-PDU (from "Responder" or "Initiator") with a matching CA;
- In the following, A-PDU "accepted" by the "Responder" is understood to mean an A-PDU "congruent" whose progressive numbering will respect the rules that will follow they will be defined step by step;
- with "n" we indicate the progressive number of the A-PDU, received and "accepted" by the “Responder”, whose value will be the highest among those of the A-PDUs already received and "accepted".

Said "Responder" must be able to handle acceptance:

- if not already done, of the "congruent" A-PDUs with progressive numbering *"q"* whose value is *nq(nI)* , with *I* configurable (note that the choice of setting I=0 is equivalent to to decide to exclude the acceptance of A-PDUs with qn);
- still to come, of the “congruent” A-PDUs with progressive numbering “q” whose value is nq(n+S), with S configurable and 1SMAX (where "MAX" is the value maximum permitted, as per point a) of point 10.7.1).

The number *"n"* constitutes the "reference" of the "sliding window" and its value takes on dynamically that of the highest progressive number *“q”* among those of the A-PDUs already received and "accepted".

The numbers "nI " and "n+S " constitute the "relative" limits, respectively lower and upper, of the same window.

The numbers *I* and *S* constitute the "absolute" limits, respectively lower and upper, of the same window.

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Finally, an A-PDU received by the "Responder" must be considered "valid", and therefore also be accepted, if it possesses the following requirements:

- its CA must be congruent;
- its progressive numbering must be included within the "relative" limits of the "sliding window";

- must be the first to be received with the aforementioned progressive numbering.

For the A-PDUs received by the “Responder” which will instead be considered “invalid”, namely:

- with progressive numbering outside the "relative" limits;
- with progressive numberings within the "relative" limits, but with inconsistent CAs;
- with progressive numbers included in the "relative" limits, with congruent CAs, but each already preceded by a "valid" A-PDU with identical numbering;

the acceptance criteria used by the "Responder" will be characterised by rules and procedures applied as described above.

10.7.3 Criteria and procedures for responding to A-PDUs received from the "Responder"

Criteria and procedures must be adopted by the "Responder" in order to manage the most "efficient" way possible (also from the point of view of saving on energy consumption batteries) responses to incoming A-PDUs as "valid" and "invalid".

Taking into account the definition of "validity" or otherwise of an A-PDU received from the "Responder" and that:

- an A-PDU received by the "Responder" can be considered as "associable" to the one consecutively received previously if their numbers are identical progressive; the received A-PDU considered "associable" may not even be processed to verify its “congruence”;
- for "spontaneous" "exercise" A-PDUs, the "Responder" may not provide any answer,
- for "broadcast" or "multicast" A-PDUs, the "Responder" does not have to provide any answer;

the following requirements must be met:

- The "Responder" responds to "invalid" or "bindable" commands from the "Initiator" only a limited and configurable number of times (for example: three); if the Initiator
- command is "associable" of "read" or a "spontaneous" one, the response is the one relating to the request (in this way the response is always updated);
- if the command is "invalid" or "bindable" write (or request for execution of a function), the command must not be executed but must be provided response which must contain an error code and the information that allow the "Initiator" to identify the reason for the failure to execute (for example: response with code indicating "write command already executed", reporting the value of the current message numerator).

An example of response criteria and procedures is shown in Table 3, represented in form of "states", "events" and "actions", applicable for all A-PDU types except quelle "spontanee" e quelle "broadcast" o "multicast".

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Table 3 **Example of response criteria and procedures**

State Current	Events	Actions	State Next
START	AMBER	[AV]a0; I_AC	A0
	R_NAME	[NAME]b0; I_AN	B0
A0	AMBER	[AV]a0; I_AC	A0
	R_ ANV (with ANV "associable" to [AV]a0)	[NAME]a1 ; I_AC	A1
	R_ANV (with ANV not "associable" to [AV]a0)	[NAME]b0; I_AN	B0
A1	AMBER	[AV]a0; I_AC	A0
	R_ ANV (with ANV "associatable" to [ANV]a1)	[NAME]a2; I_AC	A2
	R_ANV (with ANV not "associable" to [ANV]a1)	[NAME]b0 ; I_AN	B0
A2	AMBER	[AV]a0; I_AC	A0
	R_ ANV (with ANV "associable" with [ANV]a2)	[ANV]a2; No Sending	A2
	R_ANV (with ANV not "associable" with [ANV]a2)	[NAME]b0; I_AN	B0
B0	AMBER	[AV]a0; I_AC	A0
	R_ANV (with consecutive ANV to [ANV]b0)	[NAME]b1; I_AN	B1
B1	AMBER	[AV]a0 ; I_AC	A0
	R_ANV (with ANV consecutive to [ANV]b1)	[ANV]b2; I_AN	B2
B2	AMBER	[AV]a0; I_AC	A0
	R_ANV (with consecutive ANV to [ANV]b2)	[ANV]b2; No Sending	B2

NOTE

R Reception.

I Sending.

OF: A-PDU Valid.

ANV: A-PDU Invalid.

AND: A-PDU "Congruent" with what is requested in the received A-PDU.

AN: A-PDU Notification of the "invalidity" of the received A-PDU.

[XXX]an: "action" representation of the A-PDU (AV or ANV) last received (reported in "event"), marked by the subscripts "an" (a= a, b ; =0, 1, 2) to make it reference identifiable when called in "events" immediately following the aforementioned "action".

A0, A1, A2: states connected to "events" of reception of consecutive and non-consecutive AVs and ANVs and "associable" to the AVs.

B0, B1, B2: states linked to "events" of receiving non-"associable" ANVs and ANVs consecutive to non-"associable" ones.

10.8 Specificity of Translator encryption keys

10.8.1 General information

The Translator, unlike the Concentrator, does not keep a copy of the encryption keys. service and/or operation and/or installation/maintenance and/or user, of its "affiliates".

The Translator does not perform any processing of the A-PDUs related to the messages end-to-end exchanged between SAC or Concentrator and GdM.

The Translator for the functions related to network management only can manage a further encryption/authentication key called "network key" with the following properties:

- has an unlimited lifespan;

- can only be reconfigured with appropriate service messages sent by the SAC;

  - if it has a null value it is disabled;

  - allows you to carry out synchronization activities of the affiliates' clocks in the mode indicated in point 10.8.2;

  - allows you to carry out transfer activities of new software intended for affiliates (see point 11);

The "network" key in GdMs and Repeaters affiliated with a Translator must be configured/reconfigured by the SAC.

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The “network” key must remain stored in the Operator's central system.

A-PDUs encrypted and authenticated with the "network" key must follow the model of numbering and acceptance indicated in point 10.7.

10.8.2 Synchronization of devices affiliated with the Translator

The clock synchronization of a GdM or Repeater affiliated to a Translator must be done to prevent the clocks of the Translator and the GdM and/or the Repeater may arise for longer than the time permitted and indicated in UNI/TS 11291-1.

Remote clock synchronization by a Translator is only allowed for GdM and the Repeaters affiliated to the Translator and "enlisted" to the SAC or who have already been recognized and taken care of by the SAC.

The GdM/Repeater is synchronized with the following independent processes mediated by the Translator:

a) the synchronization process called "light", started by the Translator which allows a limited correction of the GdM/Repeater clock and carried out independently from the Translator with commands encrypted with the **network key;**

b) the synchronization process called "strong-indirect", initiated by the SAC, which enable the GdM/Repeater, using commands encrypted with the **operating key** of the GdM/Repeater to perform a wider clock synchronization in occasion of the next light synchronization initiated by the Translator.

Under normal operating conditions the Translator autonomously performs the "light" synchronization, sending to its affiliated and enrolled GdM/Repeaters a command that meets the following requirements:

- reports the date and time with the definition of the second;
- It is encrypted and authenticated with the network key.

Upon receiving the "light" synchronization command, the GdM/Repeater must:

- accept and respond to the command only if he has already been enrolled;
- calculate the "offset" as the difference between the actual date and time and the actual date and time reported in the command;
- accept the command only if the "deviation" (in absolute value) is less than 60 s; otherwise an error condition must be returned to the SAC; only accept "deviations" for a total
- absolute value of 60 s per month; in the event that this limit is exceeded, a sum is returned to the Translator error condition.

When the SAC detects an error condition in the lightweight synchronization procedure of the GdM/Repeater, can start the "strong-indirect" synchronization procedure.

For "strong-indirect" synchronization the SAC sends an individual command to the GdM, mediated by the Translator, which must be authenticated and encrypted with the operating key of the GdM/Repeater. With the command the SAC enables the GdM/Repeater to perform the strong sync on next sync command "light" emitted by the Translator.

The GdM accepts the synchronization authorization command in any case "strong-indirect".

The "strong-indirect" synchronization enablement remains in the GdM/Repeater until next "light" synchronization command received from the Translator.

The GdM/Repeater receiving a "light" synchronization command from the Translator following a "strong-indirect" synchronization command or a command "strong-direct" synchronization, must immediately align its clock to the Translator clock contained in the light synchronization command with the synchronization rules provided for by the relevant part of the UNI/TS 11291 series.

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10.9 Operating states

The GdMs must be able to manage the following operating states:

1) factory or pre-installation *P* state: this is the state in which the GdM is found from the end of the manufacturing process until the moment of the first installation (optional); 2) *NC* state of unconfigured device: following

the first installation until the device is configured with the minimum parameters necessary for measuring and recording data (possibly also for communication); 3) *C* state or configured device state: following the configuration of the minimum parameters necessary for measuring and recording

data (possibly also for communication); 4) *M* state or maintenance device state: state assumed by the GdM following an appropriate command; in the M state, the Operator must be able to inhibit the maintenance operator from at

least some or all of the following functions: - parameter modifications (of any type); - event log reset; - software download; - modification of metrological parameters (for example gas composition); - modification of access credentials; - valve management (for GdMs that support

it).

During the initial installation phase, the GdM generates and/or activates the temporary installation/maintenance key, inhibiting, if deemed necessary, some of the functions listed above (for example: resetting the event log; changing access credentials).

During the initial installation phase, the installer uses the temporary installation/maintenance key to encrypt and authenticate messages used for the initial configuration of the device. At the end of the installation phase, the temporary key must be able to be deactivated.

The operating key is the only one used during normal operation of the GdM; the operating key should be changed periodically.

The activation of the temporary installation/maintenance key, using a command encrypted with the operating key, must be possible without the maintenance personnel being aware of the operating key in use (for example with "offline" commands).

The Translators and Repeaters must be able to manage the following operating states:

1) *NC* state of unconfigured device: following the first installation until the device is configured with the minimum parameters necessary for its operation (possibly also for communication);

2) *C* state or configured device state: following the configuration of the minimum parameters necessary for its operation;

3) *M* state or device under maintenance state: state assumed by the Translator/Repeater following an appropriate command (local or remote); in the M state, the Operator must be able to inhibit the maintenance operator from performing at least some or all of the following functions: - resetting the event log, -

downloading the software, -

changing the status of the affiliated

devices, - changing the access credentials.

During the initial installation phase, the Translator/Repeater generates and/or activates the temporary installation/ maintenance key, inhibiting, if deemed necessary, some of the functions listed above (for example: resetting the event log; changing access credentials).

During the initial installation phase, the installer uses the temporary installation/maintenance key to encrypt and authenticate messages used for the initial configuration of the device. At the end of the installation phase, the temporary key must be able to be deactivated.

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The network key is the only one used during normal operation of the Translator towards its affiliates.

The operating key is the only one to be used during normal operation of the Translator/Repeater towards the SAC.

The operating and network keys should be changed periodically.

Activation of the temporary installation/maintenance key in the Translator/Repeater, by means of an encrypted command with the relevant operating key, it must be possible to take place without the maintenance personnel are aware of the operating key in use (for example with "offline" commands).

11 PROCEDURE FOR DOWNLOADING THE GdM SOFTWARE

11.1 General information

The "download" (DL) procedure is particularly important since the DL of the Application software is the most risky operation in relation to security and also it is certainly the most "expensive" in terms of energy as it can require the transmission of a considerable amount of information.

The boundary conditions that must be taken into account when carrying out the procedure are as follows:

1) to reduce energy consumption, in point-to-multipoint networks it is permitted to carry out the Remote DL using messages transmitted in "broadcast or multicast" or intended for a set, or a subset thereof, of GdMs that have in common the same software version;

2) the transfer phase of the new software and the phase in which it can be activated. The transfer phase can be carried out with even unencrypted messages but it is necessary that the subject is authenticated responsible for the transfer. The activation phase must be subject to verification by the GdM of the completeness, integrity and authenticity of the software.

11.2 Download procedure for GdMs communicating in PM1

The procedure must also be used for class A2 GdMs that communicate in point-to-point mode with SAC.

When the software update is done remotely, the GdMs can follow the procedure identified as interoperable for downloading the software, with the following mode:

a) the SAC or the Concentrator (prompted by the SAC) enables, with an authenticated message and encrypted with operating key, all the GdMs that are affected by the DL providing also a key (Kdl) for authenticity verification: for practical reasons of not having different DL versions of the same software for each GdM, the key could be the same for all GdMs. The message reports, in addition to the parameters useful for the DL, also the effective date of the new software and the "signature" calculated on the whole code with the Kdl key;

b) the subject (SAC, Concentrator, Translator or Terminal) responsible for carrying out the DL provides for sending the code segments in a manner that is not necessarily encrypted but authenticated with the operation or maintenance or network key;

c) after receiving all the segments, the GdM, with the necessary timing, executes a checking the authenticity of the software by computing the "signature" of the received code with Kdl: if there is coherence between the received signature and the calculated one, the GdM prepares to activate the code on the indicated date otherwise reject the code received;

d) The SAC may, but does not necessarily, request the outcome of the inspection.

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If the software update must be performed locally, the procedure that ensures interoperability evolves as follows:
a) the Operator's SAC generates "self-consistent" commands intended for all GdMs being

downloaded, each encrypted and authenticated with the operating key of the relevant GdM to activate the temporary maintenance key; the SAC also generates "offline" commands intended for the individual GdMs subject to the DL, each encrypted and authenticated with the operating key of the relevant GdM, to enable the download function. Each command also provides a key (Kdl) for authenticity verification, which, for practical reasons of not having different software for each GdM, could be the same for all GdMs. The same command, in addition to the Kdl key, also reports the effective date of the new software and the

"signature" calculated on the entire code with the Kdl key; the person responsible for carrying out the DL, equipped with a Terminal, sends locally to the GdM the relevant "offline" command(s) delivered to him by the SAC for the activation of the temporary maintenance key, and for enabling the download (see previous point), finally the code segments with commands not necessarily encrypted but

b) authenticated with the maintenance key;

c) as in the case of remote download.

11.3 Download procedure for GdMs communicating in PP4

The procedure must be followed by all GdM classes with the exception of class A2 for which the procedure described in the previous point applies (download procedure for GdMs communicating in PM1).

The software must be downloaded from the Merchant's administrator profile or from other profiles authorized by the Merchant.

The software can be downloaded if the related logical and/or physical seal is deactivated.

Software download cannot be done with encrypted messages with installation/maintenance key.

Given the typical nature of point-to-point communication between the Operator and the GdM, which is considered "more secure" because the Operator and GdM remain connected for the download period, a simplified procedure is envisaged for GdMs communicating with the SAC in PP4 mode.

The Operator performs the downloading of GdMs communicating in PP4 in two phases: the first "initialization" phase, where the SAC prepares the GdM for the DL process by transferring to the GdM the information relating to the software to be transferred (for example, identifier, effective date, total number of segments to be transmitted, the redundancy code to protect the integrity of the entire code); the second "code transfer" phase, where the SAC sends successive program segments, each not necessarily encrypted but still authenticated with the AS profile's operating key.

The GdM developer must adopt appropriate tools to verify the integrity of the received software and the authenticity of the source that generated it. Any elements that allow for the verification of the authenticity of the new software are contained within the program and therefore specific to the GdM developer.

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